

FILE PERMISSIONS IN LINUX

Scenario

You are a security professional at a large organization. You mainly work with their research team. Part of your job is to ensure users on this team are authorized with the appropriate permissions. This helps keep the system secure.

Your task is to examine existing permissions on the file system. You'll need to determine if the permissions match the authorization that should be given. If they do not match, you'll need to modify the permissions to authorize the appropriate users and remove any unauthorized access.

Summary

As a security analyst supporting the research team, I was responsible for reviewing and updating file permissions on our Linux system. This was to ensure that only authorized users had access to specific directories and files. Proper access control is important for protecting sensitive research data and preventing unauthorized access.

Checked Current File Permissions

I ran the `ls -la` command to list all files and folders along with their permissions, including hidden ones. This command helped me understand which users or groups could read, write, or execute each item in the directory.

```
researcher2@e2abc6e15b47:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 10 13:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 10 14:08 ..
-rw--w---- 1 researcher2 research_team  46 Apr 10 13:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 10 13:51 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Apr 10 13:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 10 13:51 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Apr 10 13:51 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Apr 10 13:51 project_t.txt
```

Describe the Permissions String

The permissions string looks like this: `-rwxrwxrwx`.

The first character shows if it's a file (-) or directory (d).

The next three characters (rwx) are the owner's permissions: read, write, and execute.

The next set (rwx) is for the group.

The last three (rwx) are for others.

This string tells us exactly who can do what with the file.

`-rw-rw-r--` for `project_t.txt` means the owner and group can read and write, but others can only read.

`drwx--x---` for the `drafts` directory means it is a directory, the owner can read/write/execute, the group can execute.

Hidden files like `.project_x.txt` have stricter permissions (`-rw--w----`), meaning the owner has read and write access and group has write privileges.

Change file permissions

The organization determined that other shouldn't have write access to any of their files. I referred to the file permissions that I previously returned. Project_k.txt must have the write access removed for other.

```
researcher2@e2abc6e15b47:~/projects$ chmod o-w project_k.txt
researcher2@e2abc6e15b47:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 10 13:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 10 13:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_t.txt
```

The chmod command changes the permissions on files and directories. The first argument indicates what permissions should be changed, and the second argument specifies the file or directory.

Change file permissions on a hidden file

Project_x.txt was recently archived in my organization. No one should have read access to this file except the user and group.

```
researcher2@e2abc6e15b47:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@e2abc6e15b47:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 10 13:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 10 14:08 ..
-r--r----- 1 researcher2 research_team  46 Apr 10 13:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 10 13:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_k.txt
-rw----- 1 researcher2 research_team  46 Apr 10 13:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_t.txt
```

I removed write permissions from the user with u-w. Then, I removed write permissions from the group with g-w, and added read permissions to the group with g+r.

Change directory permissions

My organization only wants the researcher2 user to have access to the drafts directory and its contents.

```
researcher2@e2abc6e15b47:~/projects$ chmod g-x drafts
researcher2@e2abc6e15b47:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Apr 10 13:51 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_k.txt
-rw----- 1 researcher2 research_team  46 Apr 10 13:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 10 13:51 project_t.txt
```

Here you can see after the command only researcher2 has execute permissions. It was previously determined that the group had execute permissions, so I used the chmod command to remove them.

Summary

By using the ls -la command, I examined permission settings for all files and directories in the projects folder. I analyzed permission strings to identify security gaps and how to adjust permissions using chmod to better protect sensitive or shared content.